FALL 2026 · ASSESSMENT REPORT

Midterm 1.

Grades & question analysis

Final grades + official roster · Nov 11, 2026 · 175 exams

175	14.61 / 21	15.00 / 21	3.52	0.729
Students included	Mean score	Median score	Standard deviation	KR-20 / Cronbach's α
All answer sets resolved	69.6% average	Middle of the score distribution	Sample SD · n − 1	All 21 questions included

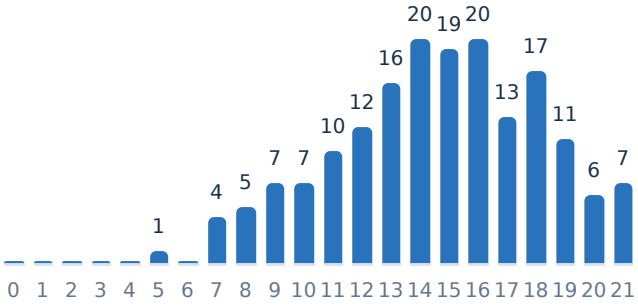
- **All answer sets resolved.** All 175 exams and all 21 questions are included. Reviewed blank answers receive zero points.

Official roster: 174 one-to-one matches · 1 auditor · 0 roster records without an exam.

CLASS PERFORMANCE

Score distribution

Range 5–21 / 21



Horizontal axis: points out of 21. Vertical bars: number of students.

25th percentile **12** Median **15** 75th percentile **17**

CONTENT AREAS

Performance by topic

Image processing	**78.1%**
Generalization and transfer	**75.6%**
Network layers	**62.4%**
Optimization and gradients	**64.5%**
Object detection	**62.3%**
Version marker	**100.0%**

Average proportion correct across the questions in each topic.

QUESTIONS TO DISCUSS

Most frequently missed

Q7 Residual connections
115 of 175 answered incorrectly
34.3% correct

Q18 Max-pooling backpropagation
105 of 175 answered incorrectly
40.0% correct

Q16 ReLU backpropagation
104 of 175 answered incorrectly
40.6% correct

These items had the lowest correct-response rates. Select an item to inspect its response distribution.

EXAM VERSIONS

Version comparison

Version	N	Mean / 21	Median	Sample SD
V0	51	14.73	15.00	3.78
V1	34	14.74	15.50	3.13
V2	37	14.84	15.00	3.47
V3	53	14.25	15.00	3.61

Descriptive statistics for each version, using the corresponding answer key.

ITEM ANALYSIS

All 21 questions

Mean is the proportion correct (0–1); SD is the sample standard deviation of each question’s 0/1 score.

Question	N	Correct	Mean (0–1)	Sample SD	Item–rest r	Blank
Q1· Box filtering ↗	175	96.0%	0.960	0.197	0.232	0
Q2· Bilateral filtering ↗	175	80.6%	0.806	0.397	0.395	0
Q3· Aliasing ↗	175	81.7%	0.817	0.388	0.214	0
Q4· Bilinear interpolation ↗	175	54.3%	0.543	0.500	0.268	1
Q5· Overfitting ↗	175	91.4%	0.914	0.281	0.150	0
Q6· Augmentation leakage ↗	175	80.0%	0.800	0.401	0.351	0
Q7· Residual connections ↗	175	34.3%	0.343	0.476	0.372	0
Q8· Convolution padding ↗	175	78.3%	0.783	0.413	0.213	0
Q9· Identity activations ↗	175	58.9%	0.589	0.494	0.330	0
Q10· Convolution output shape ↗	175	72.6%	0.726	0.447	0.343	0
Q11· Convolution parameters ↗	175	57.7%	0.577	0.495	0.343	1
Q12· Stacked convolution weights ↗	175	66.9%	0.669	0.472	0.129	0
Q13· Transfer learning ↗	175	55.4%	0.554	0.498	0.452	2
Q14· Gradient descent ↗	175	80.0%	0.800	0.401	0.245	0
Q15· Version marker: y = 2x ↗	175	100.0%	1.000	0.000	—	0
Q16· ReLU backpropagation ↗	175	40.6%	0.406	0.492	0.437	2
Q17· Chain rule ↗	175	84.0%	0.840	0.368	0.240	0
Q18· Max-pooling backpropagation ↗	175	40.0%	0.400	0.491	0.242	0
Q19· Detection pretraining ↗	175	65.7%	0.657	0.476	0.344	2
Q20· Selective search ↗	175	58.9%	0.589	0.494	0.345	0
Q21· Clearing minibatch gradients ↗	175	83.4%	0.834	0.373	0.213	0

GRADE TABLE

175 / 175 students

Student grades

Select a student to view their 21 responses and correct/incorrect results. Scores are read-only.

Record	Student · roster identity	Student ID	Version	Score / 21	Percentage
113	Nicolas Abadia Pinheiro ↗	923817422	V3	**14** / 21	66.7%
146	Mohamed Abdullrahma ↗	922858424	V0	**12** / 21	57.1%
105	Sol Abrian ↗	921991298	V2	**15** / 21	71.4%
37	Ala Alawdi ↗	925600093	V2	**14** / 21	66.7%
10	Alexander Alcaraz ↗	923926847	V3	**14** / 21	66.7%
79	Nedaa Aljalab ↗	921457745	V3	**7** / 21	33.3%
48	Brendy Alva Rodriguez ↗	923554057	V3	**8** / 21	38.1%
81	Gary Alvarado ↗	927372057	V0	**17** / 21	81.0%
147	Devi Amarsaikhan ↗	924046933	V1	**18** / 21	85.7%
51	Sasa Antunovic ↗	913502828	V1	**19** / 21	90.5%
154	Hiral Arora ↗	923782463	V2	**12** / 21	57.1%
56	Bryan Arredondo ↗	921480505	V1	**15** / 21	71.4%
92	Karina Asanbekova ↗	923403315	V3	**15** / 21	71.4%
161	Arjun Bajwa ↗	925510086	V3	**9** / 21	42.9%
54	Aiperi Baktybekova ↗	920424177	V3	**10** / 21	47.6%
53	Shaun Mathieu Banagan ↗	922001475	V3	**14** / 21	66.7%
68	Lance Barnes ↗	921135711	V1	**9** / 21	42.9%
121	Gerardo Barriga ↗	923161754	V2	**17** / 21	81.0%
17	Mina Beikzadeh Loza ↗	923378262	V3	**7** / 21	33.3%
74	Mohamad Isyraq Bin Mohamad Yusof ↗	927520207	V0	**18** / 21	85.7%
46	Millen Bisrat ↗	922201259	V1	**16** / 21	76.2%
43	Paul Boyko ↗	927267641	V0	**15** / 21	71.4%
94	Guilherme Luis Braganca ↗	923514385	V3	**21** / 21	100.0%
149	David Bui ↗	922457207	V1	**10** / 21	47.6%
3	Sebastian Bustamante Lerma ↗	921762812	V2	**12** / 21	57.1%
4	Huy Cao ↗	921147712	V3	**12** / 21	57.1%
80	Justin Chan ↗	923264059	V0	**18** / 21	85.7%
22	Wilson Chan ↗	923428882	V1	**17** / 21	81.0%
49	Matthew Chao ↗	922178722	V1	**19** / 21	90.5%
167	Jiacheng Chen ↗	927520320	V2	**13** / 21	61.9%

Record	Student · roster identity	Student ID	Version	Score / 21	Percentage
24	Maxwell Chen ↗	923689906	V0	**14** / 21	66.7%
87	Ruiqi Chen ↗	923461221	V3	**16** / 21	76.2%
86	Aunish Chhetri ↗	921997328	V2	**8** / 21	38.1%
120	Robert Choi ↗	922786622	V3	**9** / 21	42.9%
14	Parnita Chowtoori ↗	921803768	V3	**15** / 21	71.4%
82	Maitreya Das ↗	922070037	V0	**18** / 21	85.7%
16	Charitha Dendhi ↗	922326998	V0	**14** / 21	66.7%
110	Nidhi Deshmukh ↗	922439856	V3	**21** / 21	100.0%
26	Puja Devarasetty ↗	922007369	V2	**14** / 21	66.7%
76	Ralph Doria ↗	925263838	V3	**18** / 21	85.7%
96	Janna Fasheh ↗	922141827	V1	**16** / 21	76.2%
84	Elena Shu Xian Foong ↗	923202862	V2	**14** / 21	66.7%
28	Soham Gaikwad ↗	923238357	V2	**7** / 21	33.3%
73	Pranaya Rao Ganta ↗	922305346	V0	**12** / 21	57.1%
173	Matteo Godinez ↗	923167554	V0	**19** / 21	90.5%
6	Xavier Gonzalez ↗	922255250	V1	**10** / 21	47.6%
133	Erika Gosti ↗	922370423	V3	**19** / 21	90.5%
97	Samuel Greene ↗	922651676	V0	**19** / 21	90.5%
111	Cole Hanstad ↗	922562189	V2	**18** / 21	85.7%
50	Yida He ↗	927520249	V3	**16** / 21	76.2%
63	Adam Hetherwick ↗	921436818	V3	**15** / 21	71.4%
123	Logan Hillyer ↗	924812206	V1	**18** / 21	85.7%
30	Edmund Hood ↗	923505556	V2	**15** / 21	71.4%
152	Arthur Hsueh ↗	922681552	V0	**20** / 21	95.2%
47	Ilyas Hussaini ↗	922704905	V2	**20** / 21	95.2%
139	Noah Im ↗	923109389	V3	**11** / 21	52.4%
19	Sanjith Iype ↗	922364264	V2	**16** / 21	76.2%
168	Achintya Jaimini ↗	921802112	V0	**9** / 21	42.9%
52	Yujie Jiang ↗	922456224	V3	**17** / 21	81.0%
164	Faraz Kabiri ↗	921913807	V2	**20** / 21	95.2%
138	Adam Kainikara ↗	923742416	V1	**14** / 21	66.7%
27	Aryan Kapoor ↗	923410078	V1	**15** / 21	71.4%
18	Pranav Kapoor ↗	922025736	V0	**14** / 21	66.7%
7	Palina Karzhenka ↗	922262887	V3	**16** / 21	76.2%
58	Prabhjot Kaur ↗	923965688	V0	**13** / 21	61.9%

Record	Student · roster identity	Student ID	Version	Score / 21	Percentage
60	Eric Kong ↗	920794218	V0	**15** / 21	71.4%
129	Omkar Kulkarni ↗	923672007	V2	**14** / 21	66.7%
175	Rishi Kumar ↗	922196416	V0	**18** / 21	85.7%
131	Danny Kwong ↗	923522885	V1	**10** / 21	47.6%
169	Angelina Lai ↗	922549186	V0	**17** / 21	81.0%
25	Aseel Ahmed Lakadia ↗	923291242	V2	**16** / 21	76.2%
107	Yana Lapteva ↗	923809595	V3	**21** / 21	100.0%
1	Kevin Le ↗	922644858	V2	**19** / 21	90.5%
119	Nathan Le ↗	923983770	V3	**13** / 21	61.9%
155	Justin Li ↗	921903273	V3	**13** / 21	61.9%
61	Runling Li ↗	922229105	V0	**20** / 21	95.2%
67	Siyu Li ↗	921916435	V1	**16** / 21	76.2%
163	Xinyi Li ↗	922481338	V1	**16** / 21	76.2%
160	Yifan Li ↗	922155525	V1	**16** / 21	76.2%
132	Yao Lin ↗	923372019	V1	**14** / 21	66.7%
114	Fanyou Liu ↗	927550484	V0	**13** / 21	61.9%
100	Jodi Liu ↗	921810515	V0	**11** / 21	52.4%
44	Nathon Lo ↗	923262297	V3	**13** / 21	61.9%
151	Anant Madhok ↗	923002907	V2	**11** / 21	52.4%
23	Nour Makhoul ↗	926336049	V0	**18** / 21	85.7%
153	Leonardo Manieri ↗	927520265	V3	**18** / 21	85.7%
41	Himani Manjunath ↗	923180010	V0	**11** / 21	52.4%
127	Tarini Maram ↗	922038642	V2	**17** / 21	81.0%
106	Ian Marwong ↗	922205543	V2	**14** / 21	66.7%
91	Cesar Mestraitua Tarrat ↗	927513489	V0	**14** / 21	66.7%
39	Doga Mujde ↗	927539379	V1	**14** / 21	66.7%
11	Maha Muthupalaniappan ↗	922393264	V2	**18** / 21	85.7%
29	Sahana Narayan ↗	921988667	V2	**13** / 21	61.9%
103	Vivek Narayana ↗	921939098	V2	**15** / 21	71.4%
78	Subhan Nasir ↗	923865034	V3	**15** / 21	71.4%
98	Martin Ng ↗	927520223	V3	**15** / 21	71.4%
116	Hai Dang Nguyen ↗	922382602	V3	**16** / 21	76.2%
118	Xunye Nian ↗	917453193	V0	**9** / 21	42.9%
122	Clara Ogus ↗	921778576	V0	**19** / 21	90.5%
112	Luis Olea ↗	923189639	V0	**13** / 21	61.9%

Record	Student · roster identity	Student ID	Version	Score / 21	Percentage
66	Millie Orellana ↗	925529666	V3	**14** / 21	66.7%
158	Edriss Osman ↗	921058542	V2	**10** / 21	47.6%
83	Eric Pacini ↗	924642722	V3	**13** / 21	61.9%
142	Bhavya Paleru ↗	923401825	V3	**15** / 21	71.4%
90	Srihita Panati ↗	921940320	V3	**8** / 21	38.1%
126	Sathvik Parasa ↗	922076112	V2	**12** / 21	57.1%
135	Jason Peng ↗	922610646	V3	**17** / 21	81.0%
165	Daniel Pinto ↗	924685226	V0	**11** / 21	52.4%
125	Pamela Pompa ↗	921914848	V3	**17** / 21	81.0%
109	Kabir Prasad ↗	922339684	V2	**12** / 21	57.1%
174	Rajeev Raghuram ↗	921791508	V0	**8** / 21	38.1%
71	Bryan Rahardja ↗	923002923	V0	**15** / 21	71.4%
2	Kian Raisi ↗	917154104	V3	**16** / 21	76.2%
42	Pramodya Rajapakse ↗	918482406	V1	**17** / 21	81.0%
148	Manushri Rane ↗	922220803	V1	**13** / 21	61.9%
143	Alex Ren ↗	922190726	V2	**18** / 21	85.7%
62	Jose Rios ↗	924908217	V3	**9** / 21	42.9%
21	Takuma Sakamoto ↗	927513544	V3	**13** / 21	61.9%
99	Roshni Sandhu ↗	922021183	V3	**13** / 21	61.9%
95	Vermond Sardua ↗	926979697	V0	**11** / 21	52.4%
102	Ivan Savkin ↗	921779575	V0	**11** / 21	52.4%
104	Matthew Saysamone ↗	924024560	V1	**14** / 21	66.7%
144	Kushagra Sehgal ↗	923378482	V0	**17** / 21	81.0%
13	Hardik Sharma ↗	921137544	V0	**13** / 21	61.9%
101	Trisha Sharma ↗	922236295	V3	**10** / 21	47.6%
75	Cindy Shing ↗	922013447	V2	**20** / 21	95.2%
124	Anant Kumar Singh ↗	923178080	V3	**15** / 21	71.4%
33	Atin Kumar Singh ↗	924006452	V0	**18** / 21	85.7%
159	Divyansh Singh ↗	923978196	V0	**21** / 21	100.0%
141	Aishwari Sirur ↗	923447478	V0	**15** / 21	71.4%
115	Than Sitt ↗	922982468	V1	**16** / 21	76.2%
40	Luca St Pierre ↗	922529533	V2	**9** / 21	42.9%
166	Yuxuan Sun ↗	922495340	V1	**16** / 21	76.2%
93	Nithya Sunku ↗	923186959	V3	**15** / 21	71.4%
77	Yu Bin Tevyn Tan ↗	927520197	V0	**20** / 21	95.2%

Record	Student · roster identity	Student ID	Version	Score / 21	Percentage
12	Sameer Tangirala ↗	923721123	V3	**16** / 21	76.2%
45	Manraj Thandi ↗	924914298	V1	**12** / 21	57.1%
130	Amar Thota ↗	923202590	V1	**15** / 21	71.4%
57	Sean Thurman ↗	925238504	V0	**12** / 21	57.1%
72	Bryan Tiet ↗	923919016	V3	**15** / 21	71.4%
15	Josh Tompkins-Nutt ↗	924024900	V3	**18** / 21	85.7%
9	Jai Totla ↗	923374444	V0	**14** / 21	66.7%
85	Jessica Tran ↗	922201592	V0	**5** / 21	23.8%
89	Viet-Thy Tran ↗	921883021	V0	**16** / 21	76.2%
108	Daryan Trinh ↗	923862778	V0	**21** / 21	100.0%
70	Ryan Truong ↗	924081066	V1	**14** / 21	66.7%
137	Alfred Tsang ↗	922152971	V1	**21** / 21	100.0%
170	Giorgi Tsiklauri ↗	921207898	V0	**16** / 21	76.2%
38	Reid Twitchell ↗	919264267	V0	**17** / 21	81.0%
69	Taylor Ueda ↗	922198896	V2	**12** / 21	57.1%
172	Matangi Venkatakrishnan ↗	921362733	V0	**11** / 21	52.4%
31	Ananya Venkatesan ↗	923344018	V3	**10** / 21	47.6%
117	Albert Vo ↗	924005385	V2	**19** / 21	90.5%
171	Liz Voloshin ↗	921331450	V0	**17** / 21	81.0%
162	Yiran Wang ↗	922158739	V2	**19** / 21	90.5%
88	Xian Yang Weng ↗	927520168	V2	**18** / 21	85.7%
156	Chun San Wong ↗	925238274	V1	**7** / 21	33.3%
65	Aiden Xie ↗	921865997	V0	**8** / 21	38.1%
55	Seamus Yanez ↗	927539450	V1	**16** / 21	76.2%
134	Harvey Yang ↗	923748258	V3	**21** / 21	100.0%
150	Isabelle Ye ↗	923503448	V1	**11** / 21	52.4%
157	Kartik Yedumbaka ↗	925249661	V2	**17** / 21	81.0%
20	David Yi ↗	921955849	V0	**19** / 21	90.5%
8	Nathaniel York ↗	922037740	V0	**11** / 21	52.4%
36	Becky Young ↗	923261311	V1	**13** / 21	61.9%
59	Danny Yu ↗	921852968	V3	**12** / 21	57.1%
34	Christian Zannotti ↗	923998859	V3	**19** / 21	90.5%
32	Howie Zeng ↗	926470882	V3	**16** / 21	76.2%
128	Chenxuan Zhang ↗	922184127	V0	**14** / 21	66.7%
145	Victor Zhang ↗	921858564	V1	**16** / 21	76.2%

Record	Student · roster identity	Student ID	Version	Score / 21	Percentage
35	Karen Zhao ↗	921521545	V2	**13** / 21	61.9%
136	Sicheng Zhou ↗	922474950	V1	**18** / 21	85.7%
140	Chenfei Zhu ↗	923790714	V3	**13** / 21	61.9%
5	Matthew Zhu ↗	924072208	V2	**18** / 21	85.7%
64	Francisco Chen ↗ Auditor	921681618	V3	**12** / 21	57.1%

Official roster names and SIS IDs are used for matched records. The instructor-confirmed auditor retains his source name and ID and remains included in the exam statistics.

ONE-TO-ONE RECONCILIATION

Matched to the official roster

Each matched exam belongs to one roster student, and no roster student is used twice. The auditor is labeled separately; there are no unresolved matches.

174	174	0	1
Students on roster	One-to-one exam matches	Roster records without an exam	Auditor included in statistics

Auditor

Record	Name on exam	Written / transcribed ID	Score / 21	Version
64	Francisco Chen ↗	921681618	12	V3

The instructor confirmed Francisco Chen is auditing. His score is included in the distribution and all exam statistics, and he is not expected on the enrollment roster.

Roster completion

Official roster name	SIS student ID	Grade

All 174 enrolled students now have a matched exam. There are no missing roster exams.

SCORING & PROVENANCE

How to read this report

- All 21 questions, including Q15, count equally: 1 point each, 21 total.
- Every score was checked against the updated answers and the correct exam-version key.
- All 175 answer sets are resolved. Reviewed blank answers count as incorrect.
- Identity and extra-mark flags do not affect scores, per instructor instruction.
- Where one bubble is clearly darker than an erased or lighter alternative, visual judgment selects the darker mark. Genuinely unclear rows are flagged; marks are never chosen based on the answer key.
- Official names and SIS IDs come from the supplied roster where a one-to-one match was established. No roster student receives more than one exam.
- All 175 exams remain in the exam statistics, including the instructor-confirmed auditor. All 174 enrolled students have a matched exam.
- The additional ZIP contributes 8 exams. Every bubble row was visually checked; Liz Voloshin's answers are the single circled choices on questionnaire pages 1–6 because her answer sheet is blank. Rajeev Raghuram's Q11, Q16 and Q19 are confirmed blank and earn zero.
- All 167 previously published exam scores and answer sets are preserved unchanged.
- Standard deviations use the sample formula (n − 1).
- Item–rest correlation compares each 0/1 item score with the total of the other 20 items. Constant items have no defined correlation.
- KR-20 / Cronbach's alpha uses all 21 binary question scores.
- Choice frequencies are aligned by answer text across exam versions; displayed A–H labels refer to V0.

Grades: midterm1-updated-grades.numbers · 167 original exam records

Additional exams: EXAMCOMPLETED_ECS174Lecture_#1.zip · 8 exams

Additional ZIP SHA-256:
`fa0b8c0ad608ace72086f9b01b18c3261bfdee88db8be7a2a8c985f00f5a3b8f`

Roster: ECS_174_Fall_2026_student_roster.xlsx · 174 students

Grade source SHA-256:
`7375b722e9641aaa17da7ff36919955f3f41dfd3fb93530e639f3617476d446e`

Roster source SHA-256:
`d1e15fbee55fcb2a39f6f0335513d8f71412e053b169bd3aa12f78a4ff1387cd`

This is a fixed snapshot of the instructor's updated Numbers file plus eight additional exams from the supplied ZIP. It does not read or alter the grading workspace's browser drafts.